Day 8 Revision Sheet

Oct 27, 2026 | Excel: references, SUMIFS, COUNTIFS, IF, IFERROR | Git: add, commit, push | Smooth

1. Excel references (recall of Oct 23)

- **Down changes only the row number. Across changes only the column letter. A diagonal drag does both.** $ locks whatever follows it.

Reference	Locked	Copy down	Copy across
A1	nothing	A2, A3	B1, C1
$A$1	column and row	$A$1	$A$1
$A1	column	$A2, $A3	$A1
A$1	row	A$1	B$1, C$1

- **$ goes on the part that STAYS.** `=B2*$E$1`: B2 is each row's own value (it must move), E1 is the single fixed rate (it must stay). Copied to row 4: `=B4*$E$1` = 9 * 10 = 90.
- **Offset rule:** a reference shifts by exactly the distance the formula moved. Row 2 to row 4 is +2, so E1 becomes E3 (not E4). `=$B$2*E1` in row 2 becomes `=$B$2*E3` in row 4; E3 is empty, so 5 * 0 = 0 with no error.

2. SUMIFS and COUNTIFS (recall of Oct 24 + practice)

- **SUMIF(criteria_range, criteria, sum_range):** sum range LAST. **SUMIFS(sum_range, criteria_range1, criteria1, ...):** sum range FIRST. Using the SUMIF order inside SUMIFS puts text where a range is required, so Excel rejects it or shows an error (test it in your app).
- **Per category:** `=SUMIFS($C$2:$C$11,$A$2:$A$11,D2)`. The ranges are locked (same ranges on every row); D2 is not locked (it moves down to the next category name). Totals were Pen 1620, Book 2170, Bag 1830. Cross-check: they add to 5620, the sum of the whole Amount column.
- **COUNTIFS(range1,"cond1",range2,"cond2"):** the pairs work as AND, and every range must be the same size. Rows are matched by POSITION, so `A2:A20` (19 rows) with `C2:C19` (18 rows) gives #VALUE!.
- **Read the wording for the operator:** above / over / more than -> >; at least / or more -> >=; below -> <; at most -> <=. West and above 500 gave 3 (rows 3, 6, 9). 'At least 500' gave 5, because rows 4 and 10 are exactly 500. For a range of values use the same column twice: `">=500"` and `"<=700"`.
- **Hidden trailing spaces:** 'Pen ' makes SUMIFS return 0 silently. Check with `=LEN(D2)`. This is the same bug that causes #N/A in lookups.

3. IF, AND, OR, IFERROR

- IF takes ONE test. AND / OR combine plain comparisons on single cells: `=IF(AND(E2>=40,F2>=75),"Pass","Fail")`. AND needs ALL True, OR needs ONE. (range, criteria) pairs belong to COUNTIFS / SUMIFS, not to AND.
- **Ties:** 'Grade A for 90 or more' needs >=. With >, a student with exactly 90 loses the grade.
- **IFERROR(E2/F2,0)** replaces ANY error (#DIV/0!, #N/A, #VALUE!, #REF!, #NAME?) with 0. Risk 1: the 0 is real data; AVERAGE counts it and pulls the average down, while a blank is ignored. Risk 2: it hides real mistakes (typos, hidden-space lookup failures). Better where the error is expected: `=IF(F2=0,"",E2/F2)`.

4. Git: push steps and status

- Order: `git add` (stage) -> `git commit -m "message"` (save the snapshot) -> `git push` (upload to GitHub).
- `git add` alone adds nothing ('Nothing specified, nothing added'). Give a file name or a dot: `git add filename` or `git add .`.
- **Three areas:** working tree -> staging area -> repository. After `git add`, `git status` shows 'Changes to be committed' = STAGED, not committed. After `git commit`, status is clean and `git log` shows it.
- `git status` = current state of files. `git log` = commit history only. First push: `git push origin main` after `git remote add origin <url>`; `git push -u origin main` remembers the link so later a plain `git push` works. The `!` before git is Colab only.
- Write commit messages that say what changed ('Add SUMIFS practice'), not 'third commit' three times.

5. My mistakes and the fix

What I said / did	Correct version
Copying across changes the row	Down changes the row, across changes the column. I was picturing a diagonal drag, which changes both.
Row 2 to row 4: E1 became E4	Offset is +2, so E1 becomes E3. Count the exact distance.
`">=500"` for 'above 500'	`">500"`. The boundary value is where > and >= differ. Read the wording.
COUNTIFS style inside AND: `AND(E2:E20,>=40,...)`	`AND(E2>=40,F2>=75)`: plain comparisons on single cells.
COUNTIFS mismatch: 'Excel doesn't know what we want'	Rows are matched by position; row 20 of column A has no partner in column C, so it is #VALUE!.
IFERROR: 'if we remove it, average changes'	Backwards. The risk is the 0 it inserts, which AVERAGE counts, and the real errors it hides.
After git add: 'file is committed'	It is STAGED ('Changes to be committed'). Committed means it is in git log.

What I said / did	Correct version
`git add` with no path	It adds nothing. Use a file name or a dot.
Three commits named 'third commit'	Message = what changed, so the log reads like a diary.

6. How I improved today

- The SUMIFS formula had the right argument order and the right locked ranges on the first attempt, and I cross-checked the totals against the full column (5620).
- I hand-counted the West rows and found the two exactly-500 rows myself for the 'at least' case. I also sensed my IF formula had an error before being told.
- I explained git status vs git log correctly: current files vs commit history.

7. Practice to retry (no solutions)

Use a new table in A1:C7 with headers Item, Region, Qty. Rows: Shirt East 12 | Shirt West 30 | Cap West 30 | Cap East 8 | Shoe West 25 | Shoe East 30.

1. Write a SUMIFS for total Qty per Item that can be copied down. Say which parts get $ and why.
2. Predict, before using a formula: West rows with Qty at least 30, then above 30. Write both COUNTIFS and explain the difference.
3. Write an IF with OR that shows 'Restock' if Qty is below 10 OR Region is 'East', otherwise 'OK'.
4. A formula in row 3 is `=B3*$F$1`. Write it in row 7. Then write what `=$B$3*F1` becomes in row 7 and what it would calculate.
5. Why would `=SUMIFS(C2:C7,A2:A7,"Shirt ")` return 0? How do you detect the cause?
6. Put IFERROR around a division by zero. Choose 0 or blank, and explain the effect on AVERAGE.
7. In Git: change a file, run the 3 commands with a message that says what changed, and write what git status shows after add and after commit.

8. Recall questions (answer from memory)

1. What does copying down change, and what does copying across change?
2. Where does $ go, and why?
3. Which function puts the sum range first, SUMIF or SUMIFS?
4. How does COUNTIFS match rows, and what error appears when the ranges differ in size?
5. What is the difference between AND and OR? What goes inside them?
6. Give two risks of wrapping formulas in IFERROR(...,0).
7. Staged vs committed: what does git status show after add and after commit?

9. Open items

- Re-test: offset counting, AND syntax, IFERROR risks, staged vs committed (Oct 29 recall).
- Commit the Day 8 README entry as its own commit, with a message that says what changed.
- Oct 28 night tasks are next.